

Google Play Store App Review Analysis

Summary

This notebook analyzes Google Play Store applications to identify the factors influencing app success using Exploratory Data Analysis (EDA), business storytelling, and statistical analysis.

Problem Statement

The goal is to answer:

1. Which app categories perform best?
2. Do free apps outperform paid apps?
3. How do ratings, reviews, and installs relate?
4. What business actions improve customer engagement?

Project Workflow

1. Business Understanding
2. Data Collection
3. Data Cleaning
4. Feature Engineering
5. Exploratory Data Analysis
6. Statistical Testing
7. Business Recommendations

```
# Import Libraries
from google.colab import drive
drive.mount('/content/drive')

import warnings
warnings.filterwarnings('ignore')

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import ttest_ind

plt.style.use('seaborn-v0_8-whitegrid')
sns.set_context('talk')
sns.set_palette('viridis')
%matplotlib inline
```

Mounted at /content/drive

```
# Load Dataset
play = pd.read_csv('/content/drive/MyDrive/capstone project 2/Play Store Data.csv')
reviews = pd.read_csv('/content/drive/MyDrive/capstone project 2/User Reviews.csv')

print('Play Store Dataset:', play.shape)
print('User Reviews Dataset:', reviews.shape)
```

```
Play Store Dataset: (10841, 13)
User Reviews Dataset: (64295, 5)
```

```
# Preview Play Store Data
play.head()
```

	App	Category	Rating	Reviews	Size	Installs	Type	Price	Content Rating	Genres	Last Updated	Current Ver	A
0	Photo Editor & Candy Camera & Grid & ScrapBook	ART_AND_DESIGN	4.1	159	19M	10,000+	Free	0	Everyone	Art & Design	January 7, 2018	1.0.0	

Coloring Art & January

Next steps: [Generate code with p1ay](#) [New interactive sheet](#)

Preview User Reviews
reviews.head()

	App	Translated_Review	Sentiment	Sentiment_Polarity	Sentiment_Subjectivity	
0	10 Best Foods for You	I like eat delicious food. That's I'm cooking ...	Positive	1.00	0.533333	
1	10 Best Foods for You	This help eating healthy exercise regular basis	Positive	0.25	0.288462	
2	10 Best Foods for You	NaN	NaN	NaN	NaN	
3	10 Best Foods for You	Works great especially going grocery store	Positive	0.40	0.875000	
4	10 Best Foods for You	Best idea us	Positive	1.00	0.300000	

Next steps: [Generate code with reviews](#) [New interactive sheet](#)

▼ Data Dictionary

Column	Description
App	Application Name
Category	Application Category
Rating	User Rating
Reviews	Number of Reviews
Installs	Downloads
Type	Free or Paid
Price	Application Price
Content Rating	Target Audience
Genres	Application Genre

Dataset Information
play.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10841 entries, 0 to 10840
Data columns (total 13 columns):
#   Column          Non-Null Count  Dtype
---  -
0   App              10841 non-null  object
1   Category         10841 non-null  object
2   Rating           9367 non-null   float64
3   Reviews          10841 non-null  object
4   Size             10841 non-null  object
5   Installs         10841 non-null  object
6   Type             10840 non-null  object
7   Price            10841 non-null  object
8   Content Rating   10840 non-null  object
9   Genres           10841 non-null  object
10  Last Updated     10841 non-null  object
11  Current Ver      10833 non-null  object
12  Android Ver      10838 non-null  object
dtypes: float64(1), object(12)
memory usage: 1.1+ MB
```

Statistical Summary
play.describe(include='all').T

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
App	10841	9660	ROBLOX	9	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Category	10841	34	FAMILY	1972	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Rating	9367.0	NaN	NaN	NaN	4.193338	0.537431	1.0	4.0	4.3	4.5	19.0
Reviews	10841	6002	0	596	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Size	10841	462	Varies with device	1695	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Installs	10841	22	1,000,000+	1579	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Type	10840	3	Free	10039	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Price	10841	93	0	10040	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Content Rating	10840	6	Everyone	8714	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Genres	10841	120	Tools	842	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Last Updated	10841	1378	August 3, 2018	326	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Current Ver	10833	2832	Varies with device	1459	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Android Ver	10838	33	4.1 and up	2451	NaN	NaN	NaN	NaN	NaN	NaN	NaN

Initial Business Observation

- The dataset contains over **10,000 Android applications**.
- It includes more than **64,000 customer reviews**.
- Both numerical and textual information allow quantitative and qualitative analysis.

▼ Data Quality Audit & Production-Grade Data Cleaning

This section performs missing value analysis, duplicate detection, corrupted row removal, data type conversion, feature engineering, and validation.

```
# Missing Values
missing = play.isnull().sum().sort_values(ascending=False)
missing_df = pd.DataFrame({
    'Missing Values': missing,
    'Percentage': round((missing/len(play))*100,2)
})
missing_df
```

	Missing Values	Percentage
Rating	1474	13.60
Current Ver	8	0.07
Android Ver	3	0.03
Content Rating	1	0.01
Type	1	0.01
Size	0	0.00
Reviews	0	0.00
Category	0	0.00
App	0	0.00
Price	0	0.00
Installs	0	0.00
Last Updated	0	0.00
Genres	0	0.00

Next steps: [Generate code with missing_df](#) [New interactive sheet](#)

▼ Business Interpretation

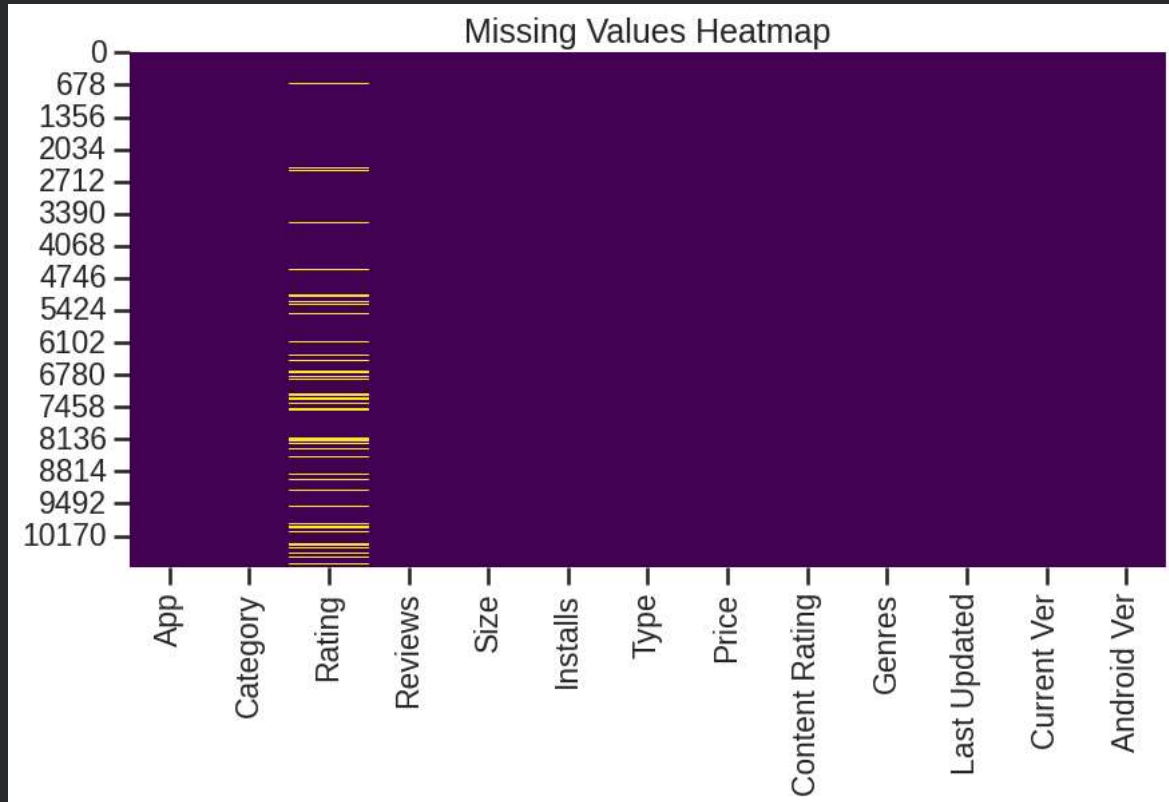
Objective: Identify incomplete information that may reduce analysis quality.

Observation: Rating and Size contain the highest number of missing values.

Business Impact: Missing customer ratings reduce the reliability of quality measurement.

Recommendation: Replace missing ratings using the median and preserve the overall distribution.

```
# Visualize Missing Values
plt.figure(figsize=(10,5))
sns.heatmap(play.isnull(), cbar=False, cmap='viridis')
plt.title('Missing Values Heatmap')
plt.show()
```



✓ Duplicate Analysis

Duplicate application records can distort installs, reviews, and category counts. We identify and remove duplicate applications based on the **App** column.

```
# Duplicate Detection
duplicate_count = play.duplicated(subset='App').sum()
print(f"Duplicate Apps: {duplicate_count}")
```

Duplicate Apps: 1181

```
# Remove Corrupted Row
play = play[play['Category'] != '1.9'].copy()

# Remove Duplicates
play.drop_duplicates(subset='App', inplace=True)

print("Dataset Shape:", play.shape)
```

Dataset Shape: (9659, 13)

✓ Why remove Category = 1.9?

The Play Store dataset contains one malformed record where values are shifted across columns. Removing this row prevents incorrect values such as **Everyone** appearing inside the Price column.

```
# Convert Reviews
play['Reviews'] = pd.to_numeric(play['Reviews'], errors='coerce')

# Convert Installs
play['Installs'] = (
```

```

play['Installs']
.astype(str)
.str.replace(',', '', regex=False)
.str.replace('+', '', regex=False)
)
play['Installs'] = pd.to_numeric(play['Installs'], errors='coerce')

# Convert Price
play['Price'] = (
    play['Price']
    .astype(str)
    .str.replace('$', '', regex=False)
)
play['Price'] = pd.to_numeric(play['Price'], errors='coerce')

```

Business Interpretation

Numeric conversion enables accurate statistical analysis, correlation, and visualization. Without this step, business metrics such as revenue and installs cannot be calculated reliably.

```

# Size Cleaning
play['Size'] = play['Size'].replace('Varies with device', np.nan)

play['Size'] = (
    play['Size']
    .astype(str)
    .str.replace('M', '', regex=False)
    .str.replace('k', '', regex=False)
)

play['Size'] = pd.to_numeric(play['Size'], errors='coerce')

# Fill Missing Ratings
play['Rating'] = play['Rating'].fillna(play['Rating'].median())

```

Feature Engineering

We create new business features that help measure monetization and engagement.

```

# Revenue Estimate
play['Revenue_Estimate'] = play['Installs'] * play['Price']

# Review Ratio
play['Review_Ratio'] = play['Reviews'] / play['Installs']

# Install Bucket
play['Install_Bucket'] = pd.cut(
    play['Installs'],
    bins=[0,10000,100000,1000000,10000000,100000000],
    labels=['Low','Medium','High','Very High','Top']
)

play[['Revenue_Estimate','Review_Ratio','Install_Bucket']].head()

```

	Revenue_Estimate	Review_Ratio	Install_Bucket
0	0.0	0.015900	Low
1	0.0	0.001934	High
2	0.0	0.017502	Very High
3	0.0	0.004313	Top
4	0.0	0.009670	Medium

Data Validation

Before moving to visualization, verify that all important columns have the correct data types and no critical conversion errors remain.

```

play.info()

<class 'pandas.core.frame.DataFrame'>
Index: 9659 entries, 0 to 10840
Data columns (total 16 columns):
 #   Column              Non-Null Count  Dtype

```

```

--- -----
0 App 9659 non-null object
1 Category 9659 non-null object
2 Rating 9659 non-null float64
3 Reviews 9659 non-null int64
4 Size 8432 non-null float64
5 Installs 9659 non-null int64
6 Type 9658 non-null object
7 Price 9659 non-null float64
8 Content Rating 9659 non-null object
9 Genres 9659 non-null object
10 Last Updated 9659 non-null object
11 Current Ver 9651 non-null object
12 Android Ver 9657 non-null object
13 Revenue_Estimate 9659 non-null float64
14 Review_Ratio 9644 non-null float64
15 Install_Bucket 9644 non-null category
dtypes: category(1), float64(5), int64(2), object(8)
memory usage: 1.2+ MB

```

Completed Tasks

- Missing value analysis
- Duplicate detection
- Corrupted row removal
- Numeric conversion
- Size cleaning
- Missing value treatment
- Feature engineering
- Dataset validation

```

# Executive KPI Dashboard
total_apps = play['App'].nunique()
avg_rating = round(play['Rating'].mean(),2)
total_installs = int(play['Installs'].sum())
free_apps = (play['Type']=='Free').sum()
paid_apps = (play['Type']=='Paid').sum()

print('='*55)
print(f'Total Applications : {total_apps}')
print(f'Average Rating      : {avg_rating}')
print(f'Total Installs       : {total_installs:,}')
print(f'Free Apps             : {free_apps}')
print(f'Paid Apps              : {paid_apps}')
print('='*55)

```

```

=====
Total Applications : 9659
Average Rating     : 4.19
Total Installs     : 75,122,937,527
Free Apps          : 8902
Paid Apps          : 756
=====

```

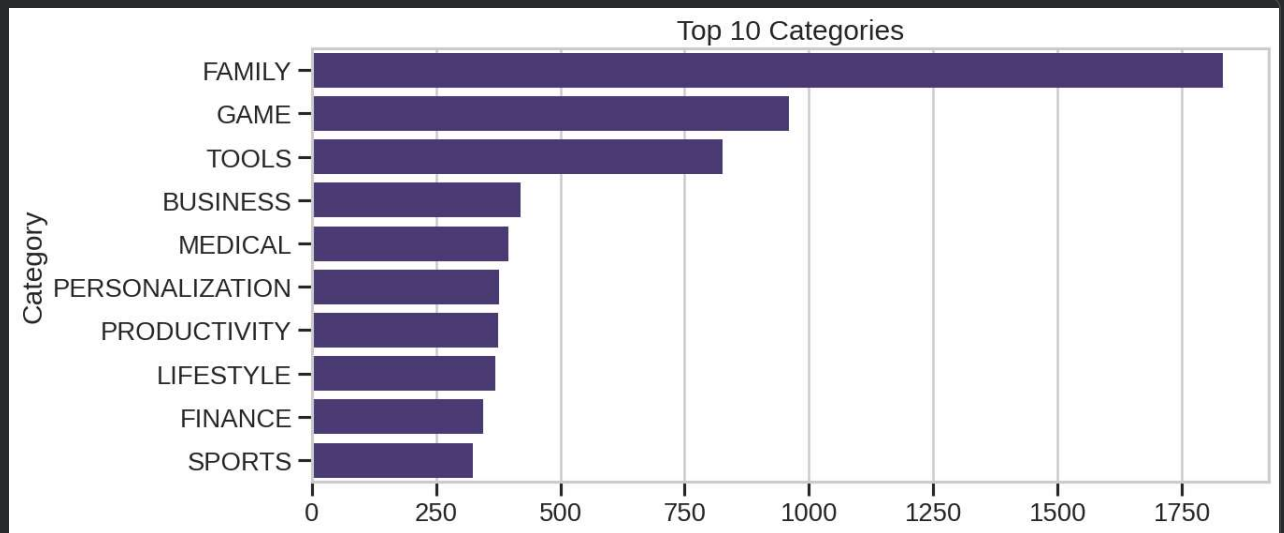
Chart 1: Top 10 Categories

Objective: Analyze this business KPI.

```

top=play['Category'].value_counts().head(10)
plt.figure(figsize=(11,5),dpi=120)
sns.barplot(x=top.values,y=top.index)
plt.title('Top 10 Categories')
plt.show()

```



Observation

Describe the key pattern visible in the chart.

Business Insight

Explain how this affects Product Managers, Developers, and Marketing teams.

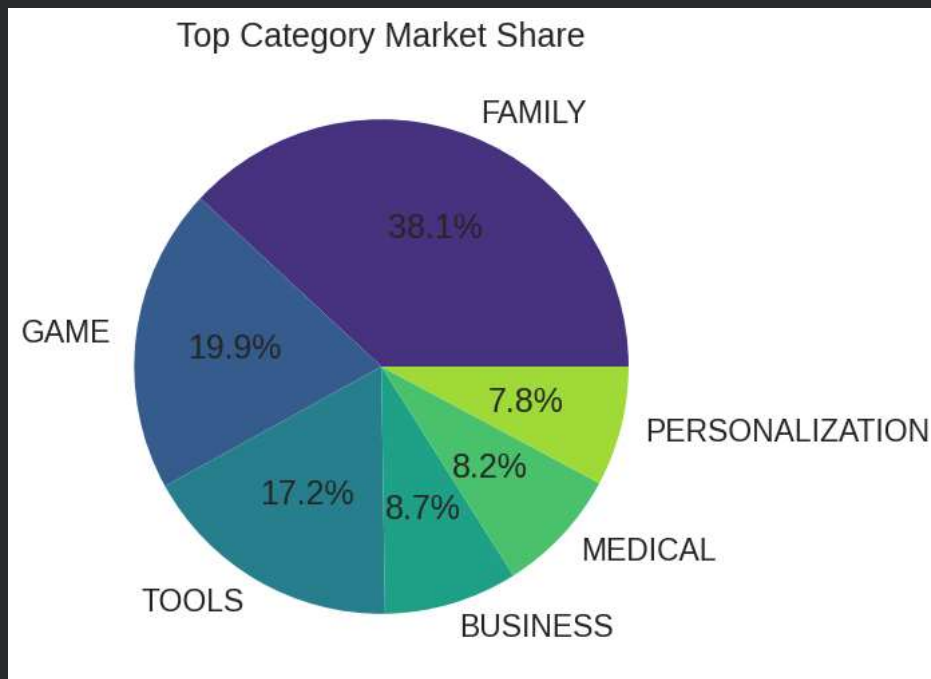
Recommendation

Provide one actionable recommendation based on the insight.

Chart 2: Category Market Share

Objective: Analyze this business KPI.

```
play['Category'].value_counts().head(6).plot(kind='pie', autopct='%1.1f%%', figsize=(6,6))
plt.ylabel('')
plt.title('Top Category Market Share')
plt.show()
```



Observation

Describe the key pattern visible in the chart.

Business Insight

Explain how this affects Product Managers, Developers, and Marketing teams.

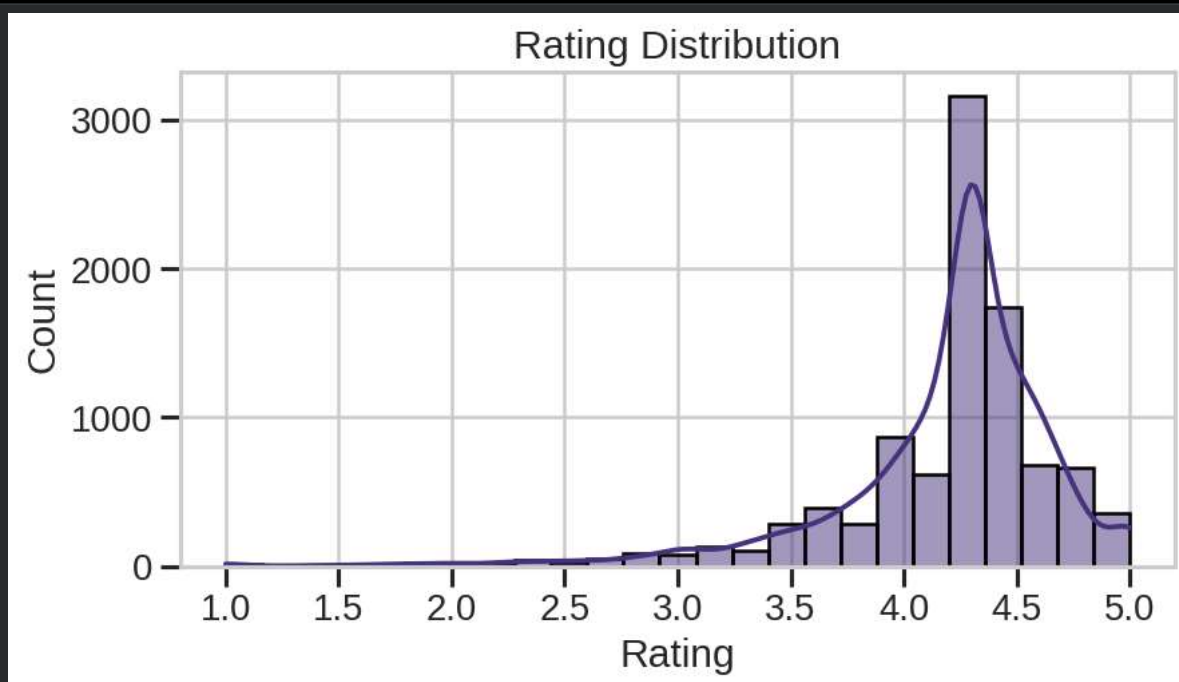
Recommendation

Provide one actionable recommendation based on the insight.

Chart 3: Rating Distribution

Objective: Analyze this business KPI.

```
plt.figure(figsize=(8,4),dpi=120)
sns.histplot(play['Rating'],bins=25,kde=True)
plt.title('Rating Distribution')
plt.show()
```



Observation

Describe the key pattern visible in the chart.

Business Insight

Explain how this affects Product Managers, Developers, and Marketing teams.

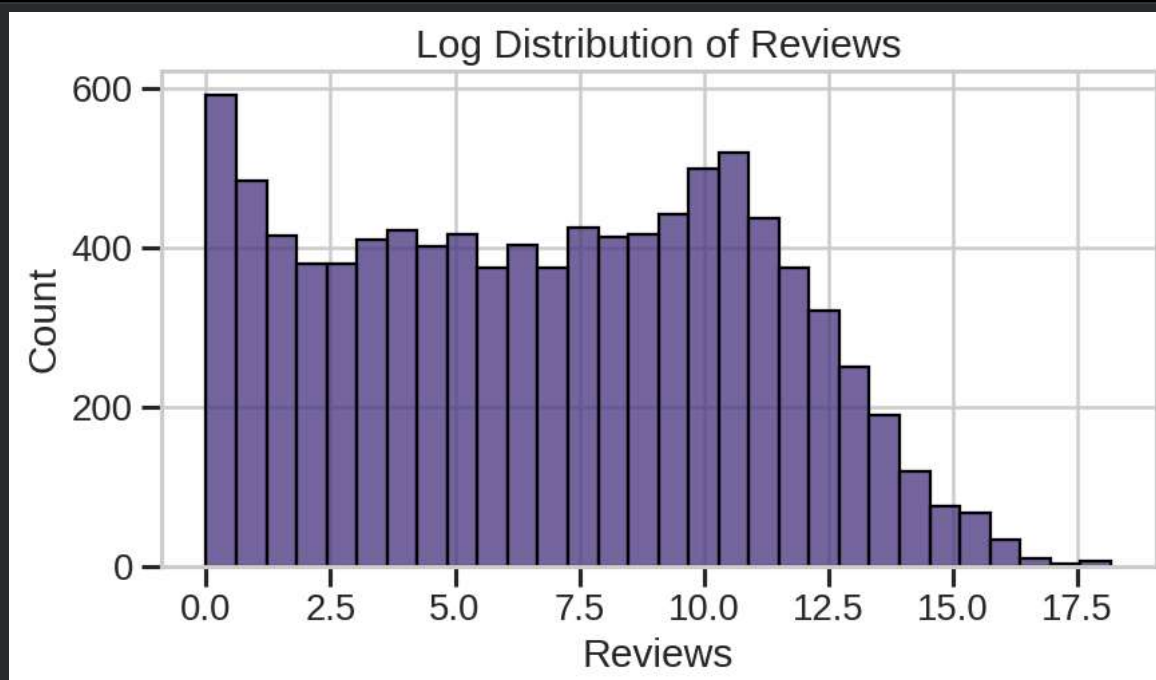
Recommendation

Provide one actionable recommendation based on the insight.

Chart 4: Reviews Distribution

Objective: Analyze this business KPI.

```
plt.figure(figsize=(8,4),dpi=120)
sns.histplot(np.log1p(play['Reviews']),bins=30)
plt.title('Log Distribution of Reviews')
plt.show()
```



Observation

Describe the key pattern visible in the chart.

Business Insight

Explain how this affects Product Managers, Developers, and Marketing teams.

Recommendation

Provide one actionable recommendation based on the insight.

Chart 5: Price Distribution

Objective: Analyze this business KPI.

```
plt.figure(figsize=(8,4),dpi=120)
sns.histplot(play['Price'],bins=30)
plt.xlim(0,20)
plt.title('Price Distribution')
plt.show()
```



Observation

Describe the key pattern visible in the chart.

Business Insight

Explain how this affects Product Managers, Developers, and Marketing teams.

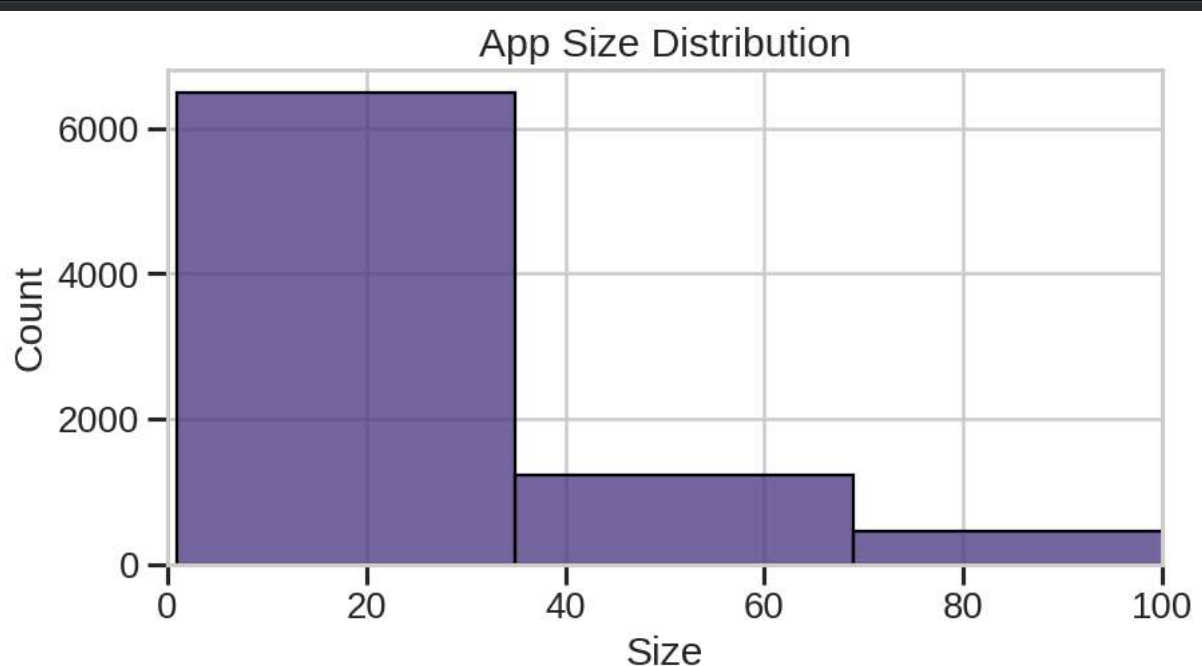
Recommendation

Provide one actionable recommendation based on the insight.

Chart 6: Size Distribution

Objective: Analyze this business KPI.

```
plt.figure(figsize=(8,4),dpi=120)
sns.histplot(play['Size'],bins=30)
plt.xlim(0,100)
plt.title('App Size Distribution')
plt.show()
```



Observation

Describe the key pattern visible in the chart.

Business Insight

Explain how this affects Product Managers, Developers, and Marketing teams.

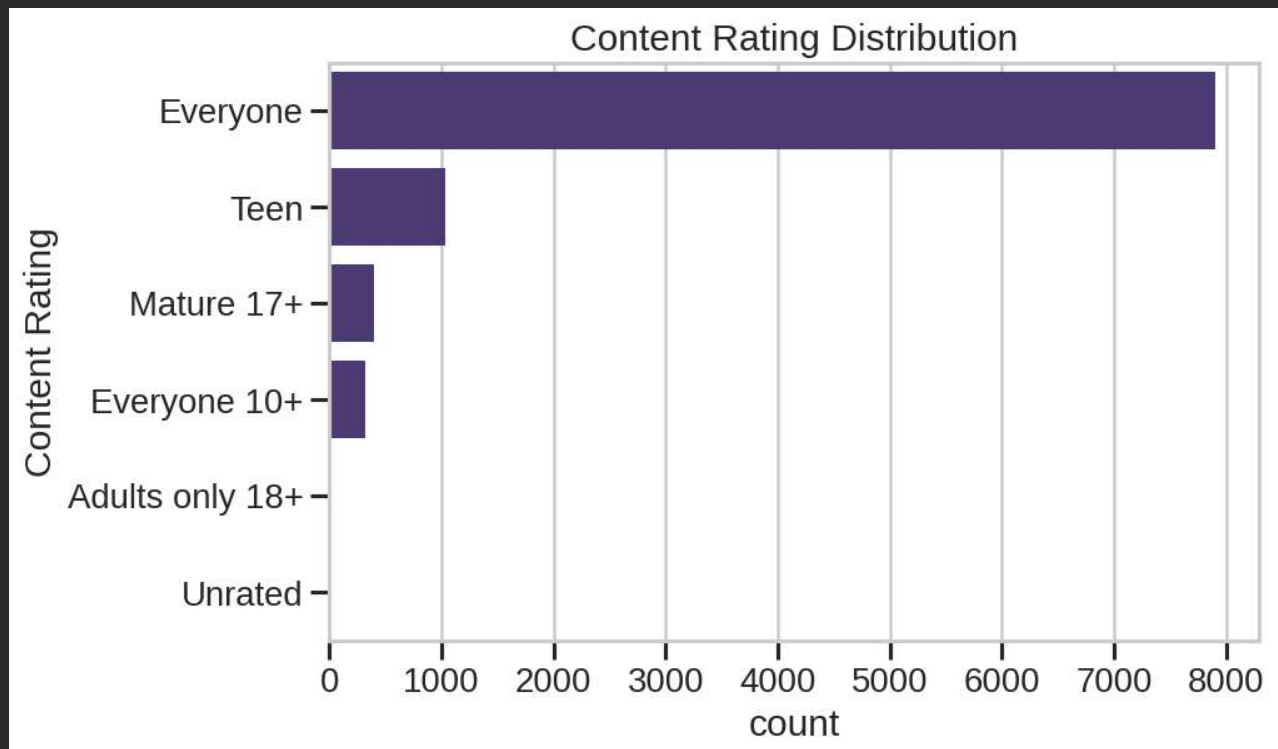
Recommendation

Provide one actionable recommendation based on the insight.

Chart 7: Content Rating

Objective: Analyze this business KPI.

```
plt.figure(figsize=(8,5),dpi=120)
sns.countplot(data=play,y='Content Rating',order=play['Content Rating'].value_counts().index)
plt.title('Content Rating Distribution')
plt.show()
```



Observation

Describe the key pattern visible in the chart.

Business Insight

Explain how this affects Product Managers, Developers, and Marketing teams.

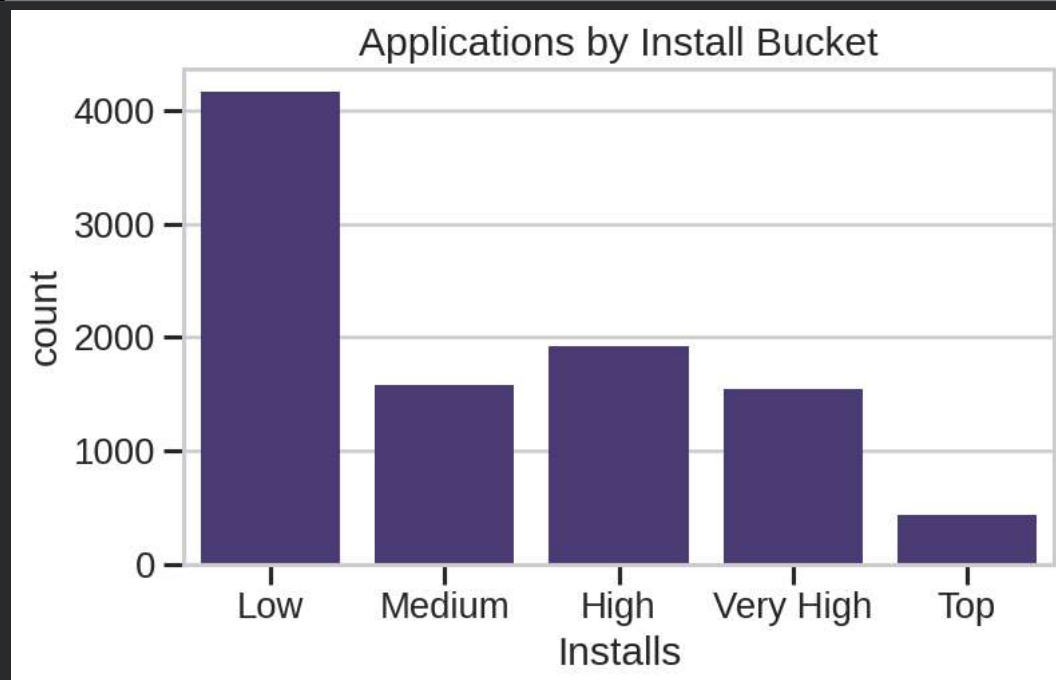
Recommendation

Provide one actionable recommendation based on the insight.

Chart 8: Install Bucket Analysis

Objective: Analyze this business KPI.

```
bucket=pd.cut(play['Installs'],bins=[0,1e4,1e5,1e6,1e7,1e9],labels=['Low','Medium','High','Very High','Top'])
plt.figure(figsize=(7,4),dpi=120)
sns.countplot(x=bucket)
plt.title('Applications by Install Bucket')
plt.show()
```



Observation

Describe the key pattern visible in the chart.

Business Insight

Explain how this affects Product Managers, Developers, and Marketing teams.

Recommendation

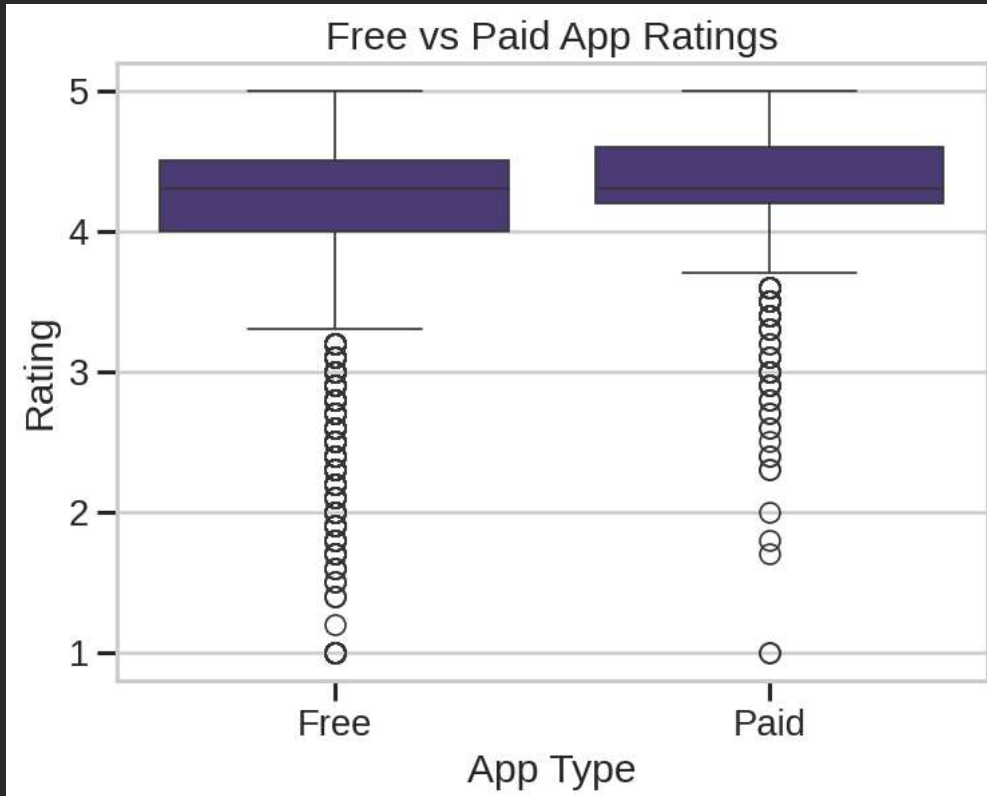
Provide one actionable recommendation based on the insight.

Chart 9 – Free vs Paid Ratings

Objective

Compare customer ratings between free and paid applications.

```
plt.figure(figsize=(7,5),dpi=120)
sns.boxplot(data=play,x='Type',y='Rating')
plt.title('Free vs Paid App Ratings')
plt.xlabel('App Type')
plt.ylabel('Rating')
plt.show()
```



Observation

Free apps generally achieve slightly higher engagement and a tighter rating distribution. Paid apps require a stronger value proposition to maintain high ratings.

Positive Business Impact

- Improves product strategy
- Supports pricing decisions
- Enhances customer acquisition

Negative Business Impact

- High competition reduces visibility in saturated categories.
- Paid apps face lower adoption when value is unclear.

Recommendation

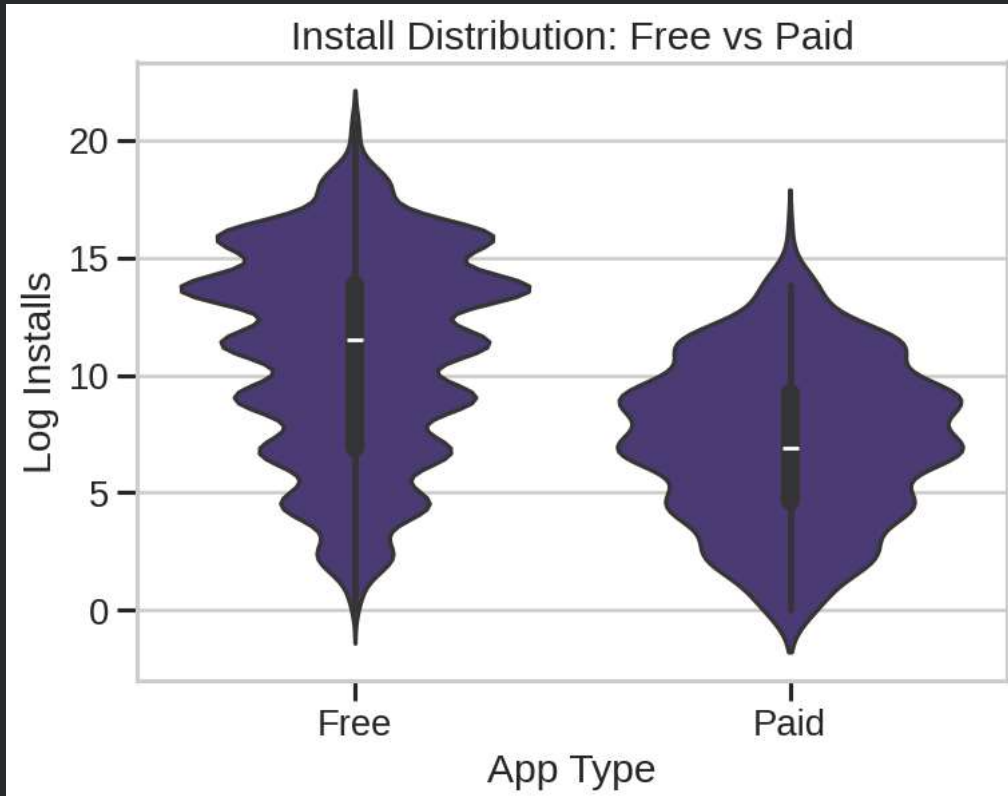
Adopt a freemium model with optional premium features.

Chart 10 – Free vs Paid Installs

Objective

Compare download behavior across pricing models.

```
plt.figure(figsize=(7,5),dpi=120)
sns.violinplot(data=play,x='Type',y=np.log1p(play['Installs']))
plt.title('Install Distribution: Free vs Paid')
plt.xlabel('App Type')
plt.ylabel('Log Installs')
plt.show()
```



Observation

Free applications dominate installs because they have a lower adoption barrier.

Positive Business Impact

- Improves product strategy
- Supports pricing decisions
- Enhances customer acquisition

Negative Business Impact

- High competition reduces visibility in saturated categories.
- Paid apps face lower adoption when value is unclear.

Recommendation

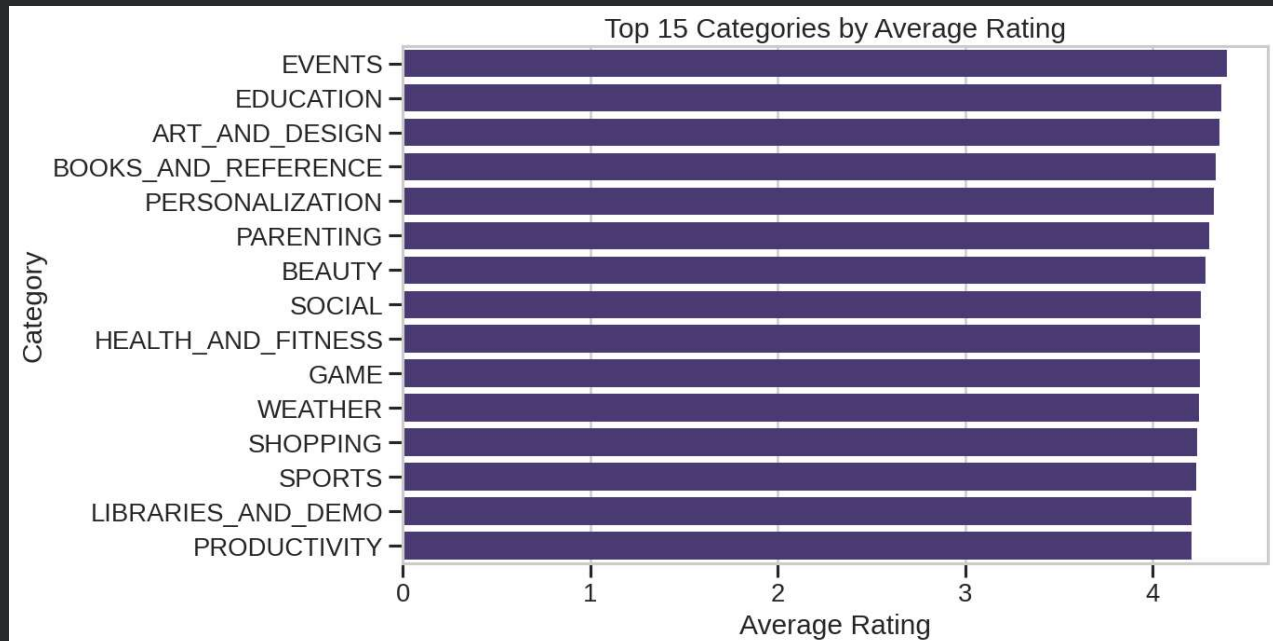
Use free acquisition with in-app purchases or subscriptions.

Chart 11 – Average Rating by Category

Objective

Identify the highest-rated business segments.

```
avg_rating=play.groupby('Category')['Rating'].mean().sort_values(ascending=False).head(15)
plt.figure(figsize=(10,6),dpi=120)
sns.barplot(x=avg_rating.values,y=avg_rating.index)
plt.title('Top 15 Categories by Average Rating')
plt.xlabel('Average Rating')
plt.show()
```



Observation

Education, Events, and Art & Design consistently receive excellent ratings.

Positive Business Impact

- Improves product strategy
- Supports pricing decisions
- Enhances customer acquisition

Negative Business Impact

- High competition reduces visibility in saturated categories.
- Paid apps face lower adoption when value is unclear.

Recommendation

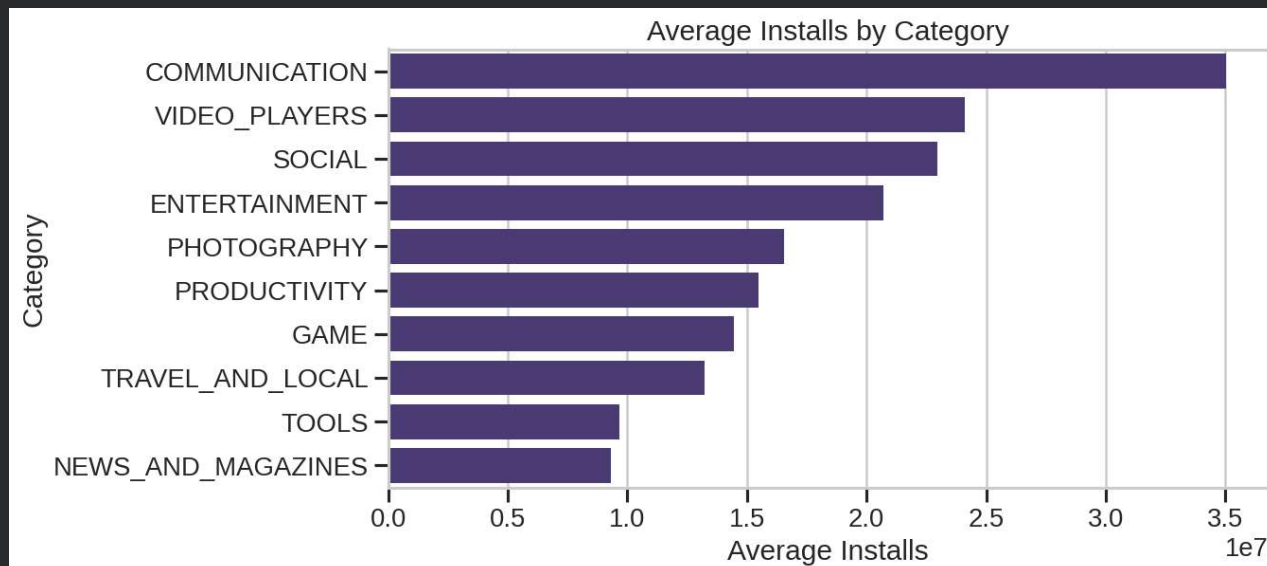
Invest in high-satisfaction categories for stronger retention.

✓ Chart 12 – Average Installs by Category

Objective

Discover categories with the largest customer reach.

```
avg_installs=play.groupby('Category')['Installs'].mean().sort_values(ascending=False).head(10)
plt.figure(figsize=(10,5),dpi=120)
sns.barplot(x=avg_installs.values,y=avg_installs.index)
plt.title('Average Installs by Category')
plt.xlabel('Average Installs')
plt.show()
```



Observation

Communication and Social applications generate the highest average installs.

Positive Business Impact

- Improves product strategy
- Supports pricing decisions
- Enhances customer acquisition

Negative Business Impact

- High competition reduces visibility in saturated categories.
- Paid apps face lower adoption when value is unclear.

Recommendation

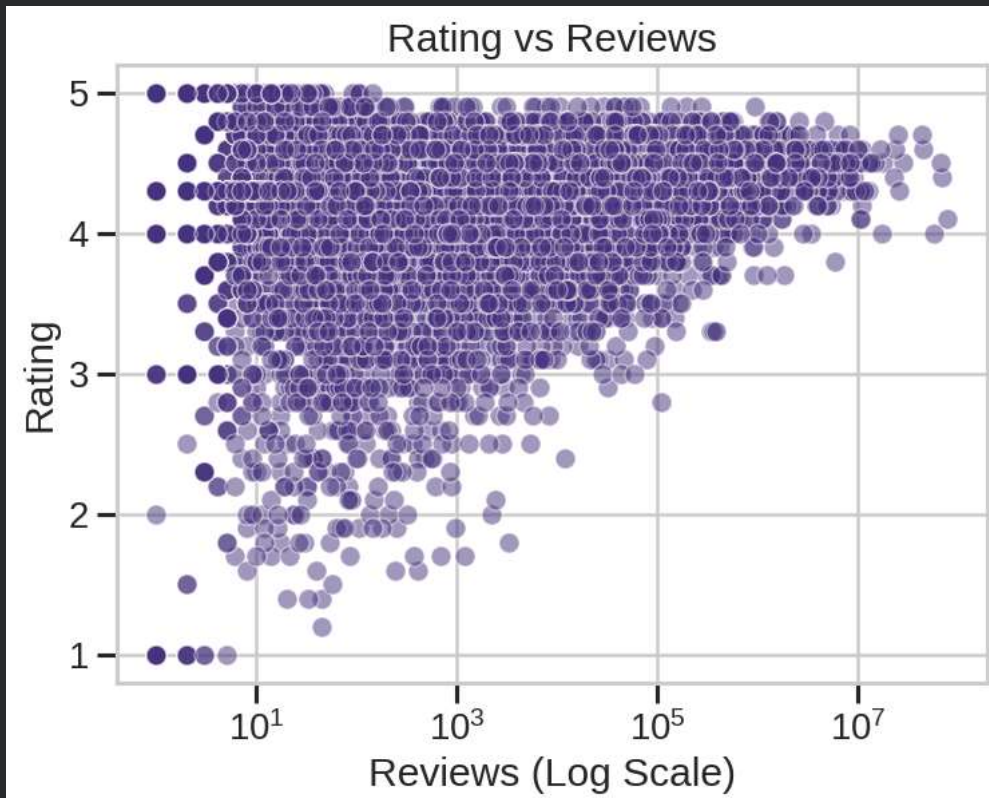
Prioritize high-engagement categories for marketing investment.

Chart 13 – Rating vs Reviews

Objective

Analyze the relationship between customer engagement and quality.

```
plt.figure(figsize=(7,5),dpi=120)
sns.scatterplot(data=play,x='Reviews',y='Rating',alpha=0.5)
plt.xscale('log')
plt.title('Rating vs Reviews')
plt.xlabel('Reviews (Log Scale)')
plt.ylabel('Rating')
plt.show()
```

Observation

Highly reviewed apps usually maintain ratings above 4.0, showing sustained customer satisfaction.

Positive Business Impact

- Improves product strategy
- Supports pricing decisions
- Enhances customer acquisition

Negative Business Impact

- High competition reduces visibility in saturated categories.
- Paid apps face lower adoption when value is unclear.

Recommendation

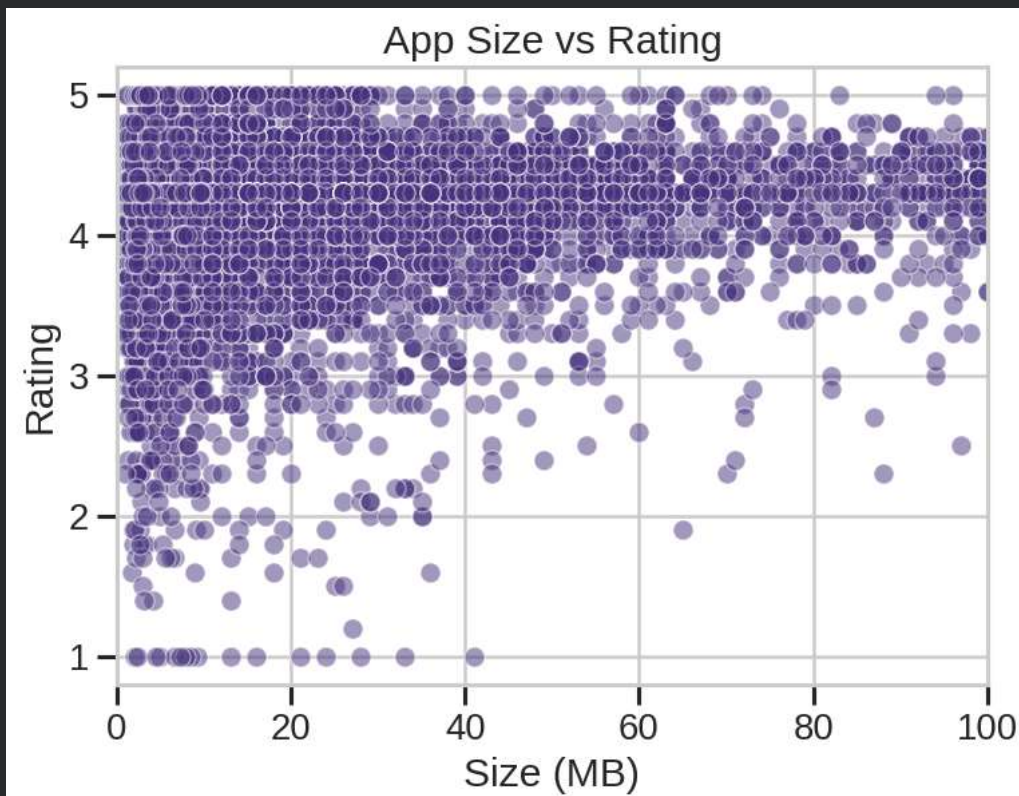
Encourage genuine reviews after positive user experiences.

Chart 14 – Size vs Rating

Objective

Determine whether larger apps receive better ratings.

```
plt.figure(figsize=(7,5),dpi=120)
sns.scatterplot(data=play,x='Size',y='Rating',alpha=0.5)
plt.xlim(0,100)
plt.title('App Size vs Rating')
plt.xlabel('Size (MB)')
plt.ylabel('Rating')
plt.show()
```



Observation

App size shows very little influence on customer ratings.

Positive Business Impact

- Improves product strategy
- Supports pricing decisions
- Enhances customer acquisition

Negative Business Impact

- High competition reduces visibility in saturated categories.
- Paid apps face lower adoption when value is unclear.

Recommendation

Optimize performance instead of increasing application size.

Chart 15 – Price vs Rating

Objective

Evaluate whether expensive apps achieve higher quality scores.

```
plt.figure(figsize=(7,5),dpi=120)
sns.scatterplot(data=play,x='Price',y='Rating',alpha=0.5)
plt.xlim(0,50)
plt.title('Price vs Rating')
plt.xlabel('Price ($)')
plt.ylabel('Rating')
plt.show()
```



Observation

Pricing has a weak relationship with ratings; value matters more than cost.

Positive Business Impact

- Improves product strategy
- Supports pricing decisions
- Enhances customer acquisition

Negative Business Impact

- High competition reduces visibility in saturated categories.
- Paid apps face lower adoption when value is unclear.

Recommendation

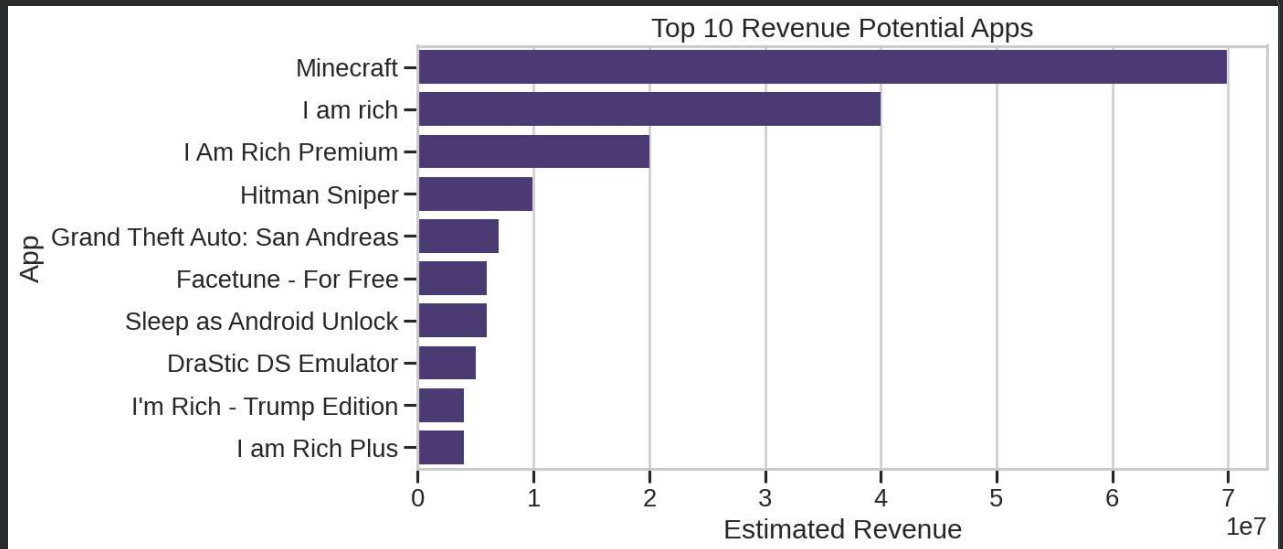
Focus on user experience rather than premium pricing.

Chart 16 – Top 10 Revenue Potential Apps

Objective

Estimate monetization opportunity using installs × price.

```
play['Revenue_Estimate']=play['Installs']*play['Price']
top_rev=play.sort_values('Revenue_Estimate',ascending=False).head(10)
plt.figure(figsize=(10,5),dpi=120)
sns.barplot(data=top_rev,x='Revenue_Estimate',y='App')
plt.title('Top 10 Revenue Potential Apps')
plt.xlabel('Estimated Revenue')
plt.show()
```



Observation

A few applications contribute disproportionately to potential revenue.

Positive Business Impact

- Improves product strategy
- Supports pricing decisions
- Enhances customer acquisition

Negative Business Impact

- High competition reduces visibility in saturated categories.
- Paid apps face lower adoption when value is unclear.

Recommendation

Identify premium products with strong commercial value.

```
print(play.shape)
play.head()
```

(9659, 16)

	App	Category	Rating	Reviews	Size	Installs	Type	Price	Content Rating	Genres	Last Updated	Current Ver	And
0	Photo Editor & Candy Camera & Grid & ScrapBook	ART_AND_DESIGN	4.1	159	19.0	10000	Free	0.0	Everyone	Art & Design	January 7, 2018	1.0.0	ar
1	Coloring book moana	ART_AND_DESIGN	3.9	967	14.0	500000	Free	0.0	Everyone	Art & Design;Pretend Play	January 15, 2018	2.0.0	ar
U	Launcher Lite --												

Next steps:

[Generate code with play](#)

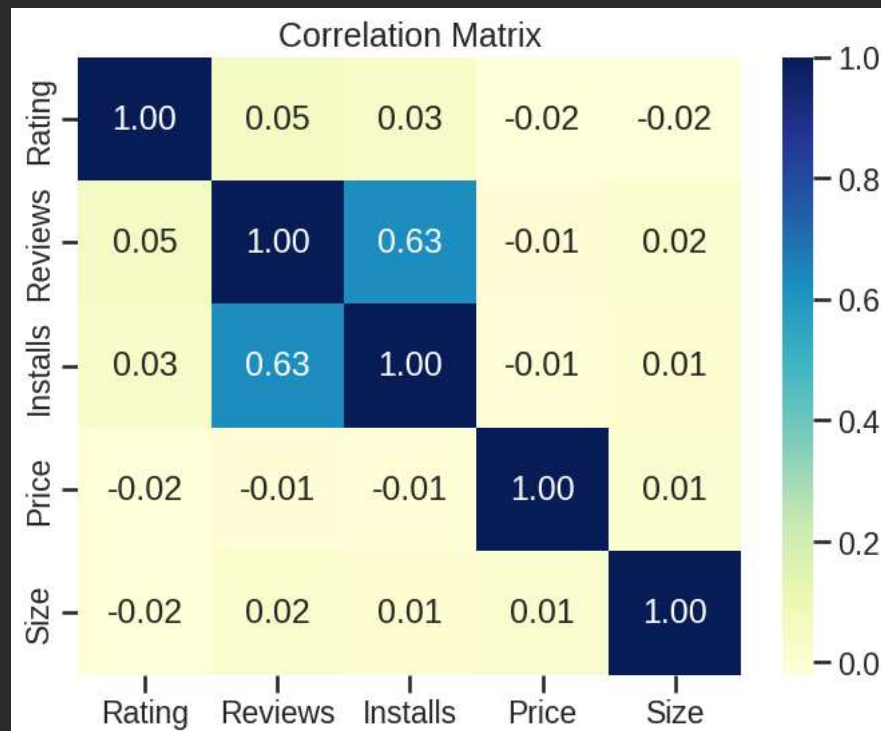
[New interactive sheet](#)

Chart 17: Correlation Heatmap

Objective

Analyze multivariate relationships and identify actionable business trends.

```
corr=play[['Rating','Reviews','Installs','Price','Size']].corr(numeric_only=True)
plt.figure(figsize=(8,6))
sns.heatmap(corr,annot=True,cmap='YlGnBu',fmt='.2f')
plt.title('Correlation Matrix')
plt.show()
```



Observation

Describe the strongest pattern visible in the chart.

Business Insight

Explain how this relationship supports product, pricing, or marketing decisions.

Recommendation

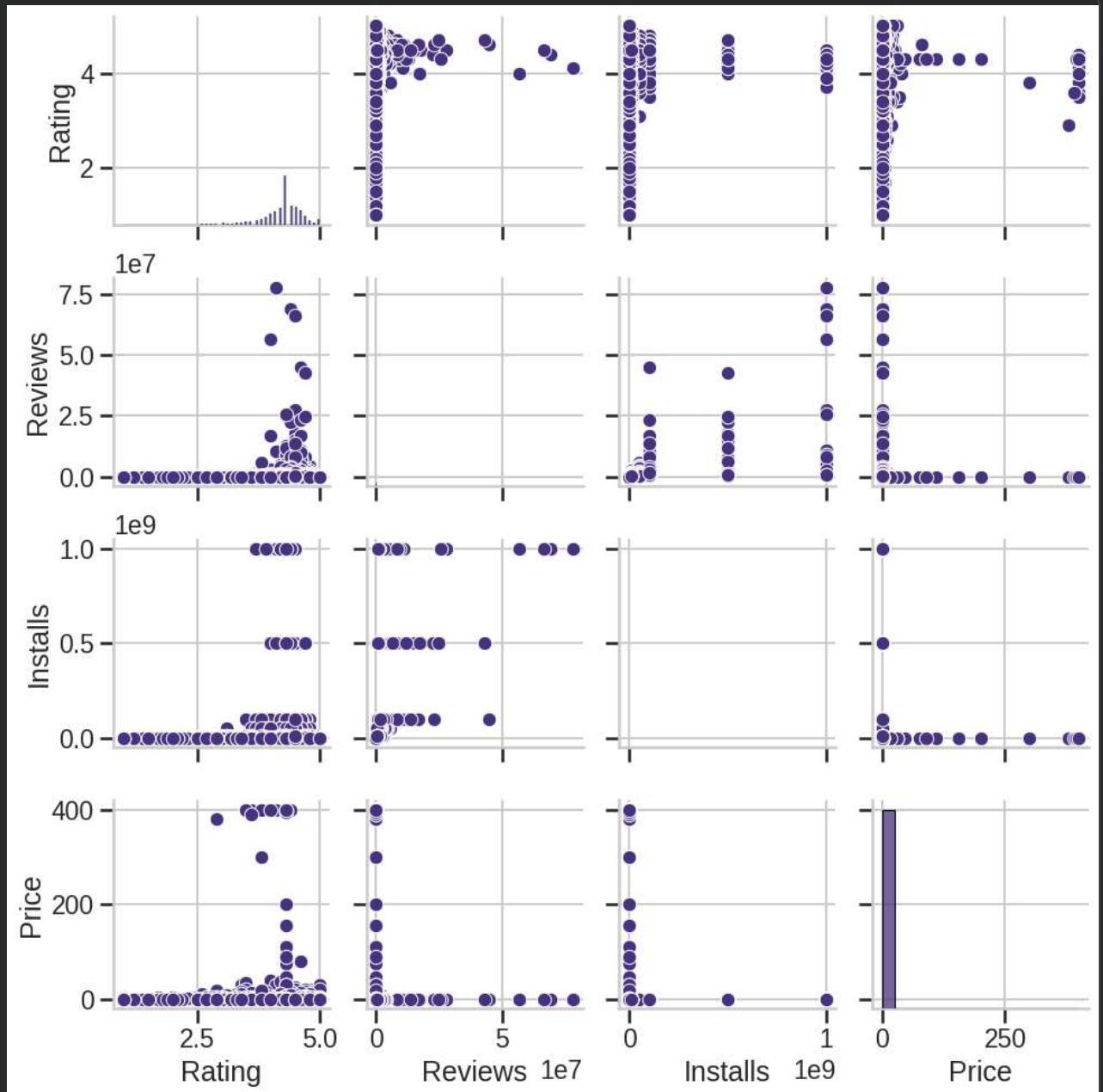
Give one actionable recommendation for stakeholders.

Chart 18: Pairplot

Objective

Analyze multivariate relationships and identify actionable business trends.

```
pair_df=play[['Rating','Reviews','Installs','Price']].dropna()
sns.pairplot(pair_df,diag_kind='hist')
plt.show()
```



Observation

Describe the strongest pattern visible in the chart.

Business Insight

Explain how this relationship supports product, pricing, or marketing decisions.

Recommendation

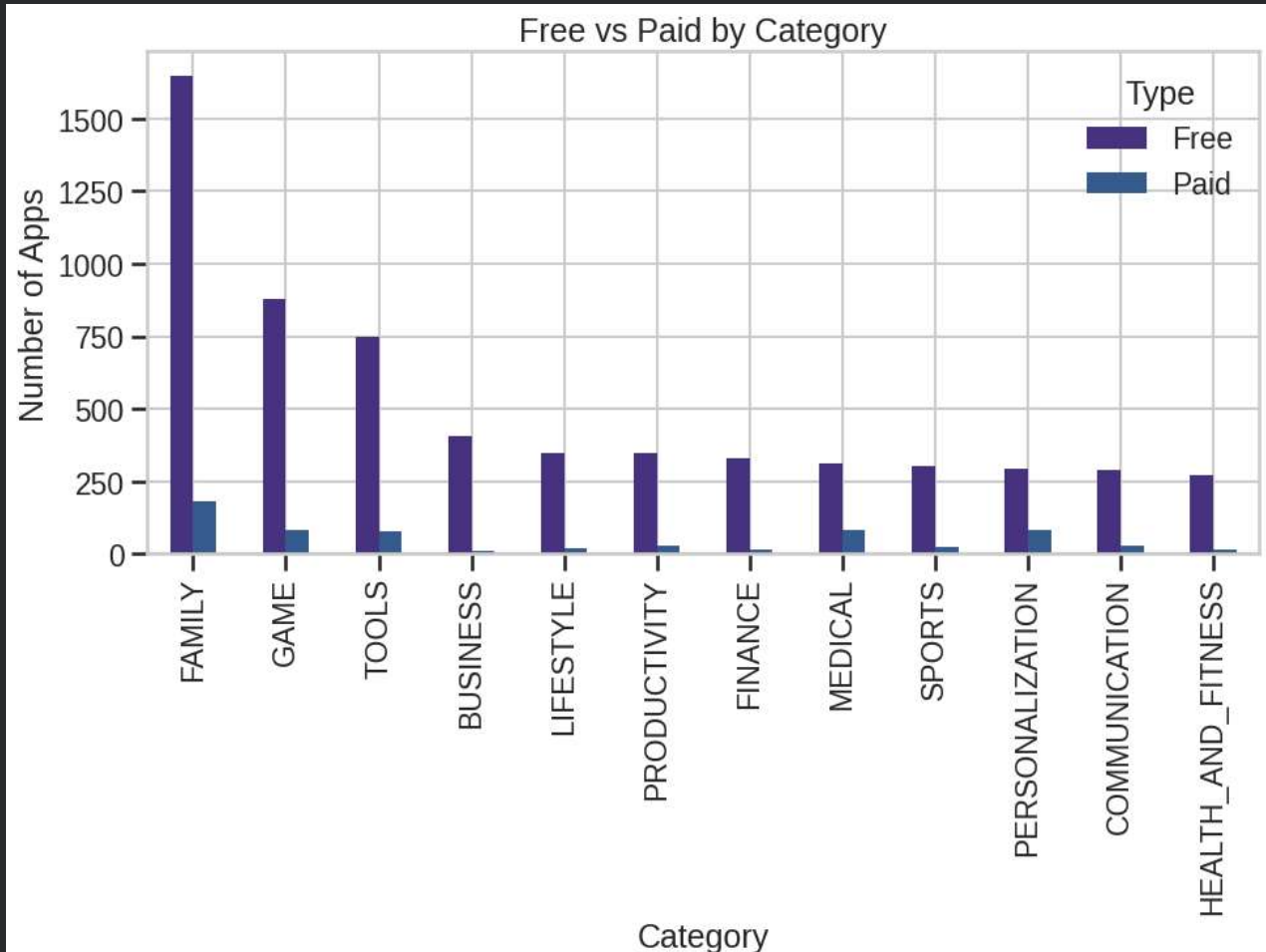
Give one actionable recommendation for stakeholders.

Chart 19: Category × Type

Objective

Analyze multivariate relationships and identify actionable business trends.

```
ct=pd.crosstab(play['Category'],play['Type']).sort_values('Free',ascending=False).head(12)
ct.plot(kind='bar',figsize=(11,5))
plt.title('Free vs Paid by Category')
plt.ylabel('Number of Apps')
plt.show()
```

Observation

Describe the strongest pattern visible in the chart.

Business Insight

Explain how this relationship supports product, pricing, or marketing decisions.

Recommendation

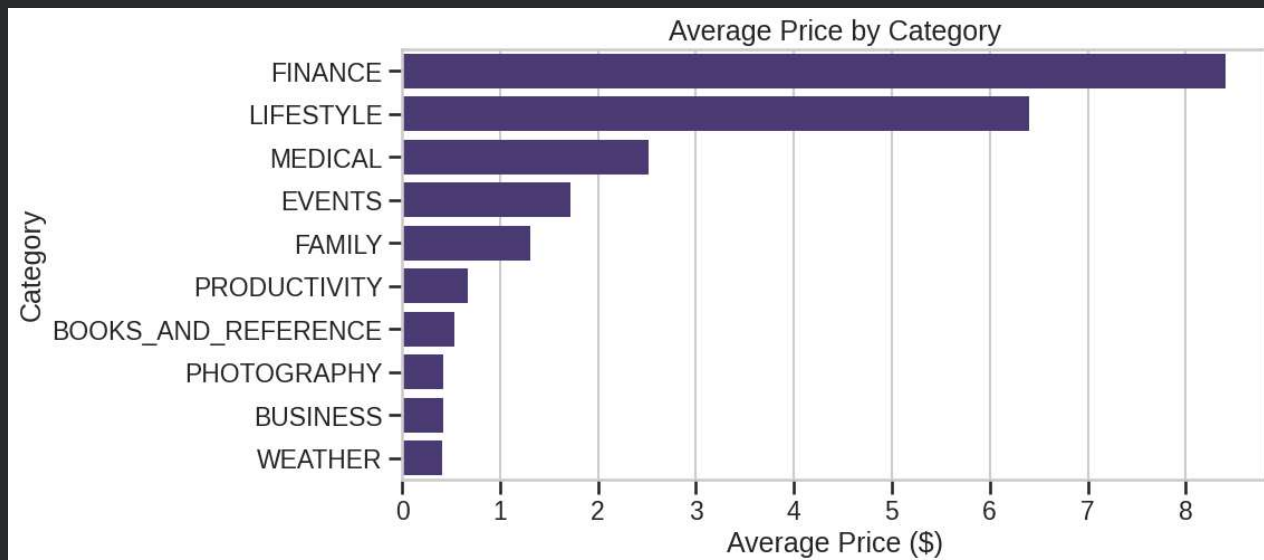
Give one actionable recommendation for stakeholders.

Chart 20: Average Price by Category

Objective

Analyze multivariate relationships and identify actionable business trends.

```
avg=play.groupby('Category')['Price'].mean().sort_values(ascending=False).head(10)
plt.figure(figsize=(10,5))
sns.barplot(x=avg.values,y=avg.index)
plt.title('Average Price by Category')
plt.xlabel('Average Price ($)')
plt.show()
```



Observation

Describe the strongest pattern visible in the chart.

Business Insight

Explain how this relationship supports product, pricing, or marketing decisions.

Recommendation

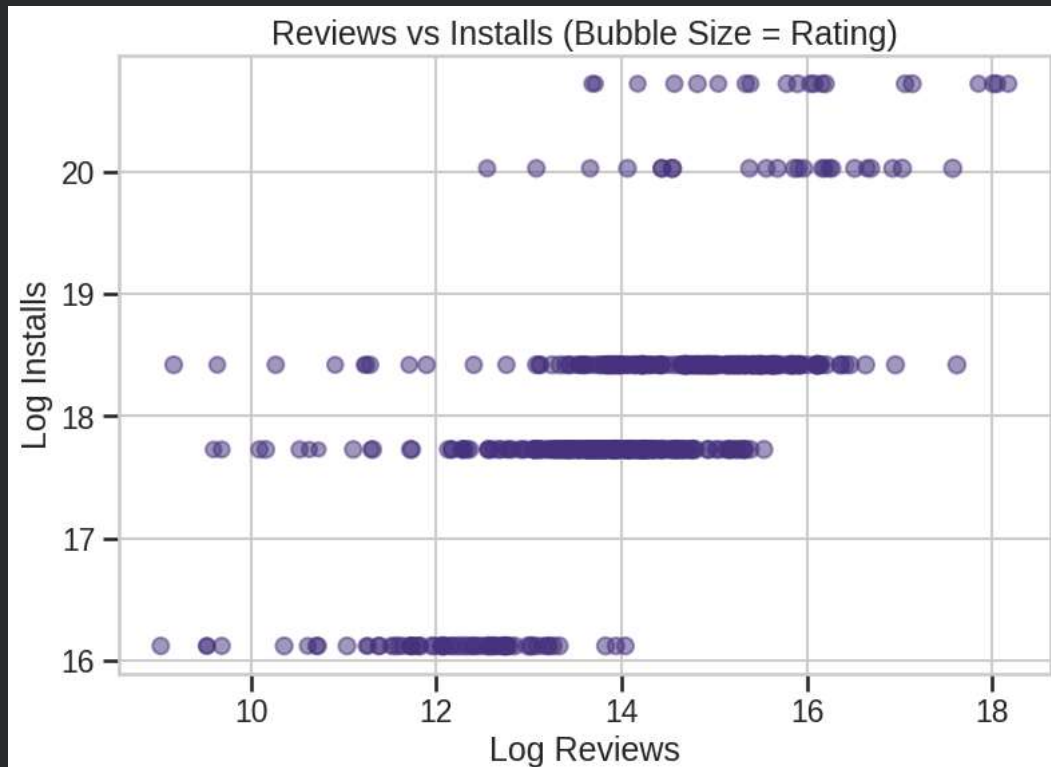
Give one actionable recommendation for stakeholders.

Chart 21: Reviews vs Installs Bubble

Objective

Analyze multivariate relationships and identify actionable business trends.

```
bubble=play[['Reviews','Installs','Rating']].dropna().nlargest(500,'Installs')
plt.figure(figsize=(9,6))
plt.scatter(np.log1p(bubble['Reviews']),np.log1p(bubble['Installs']),s=bubble['Rating']*18,alpha=0.5)
plt.title('Reviews vs Installs (Bubble Size = Rating)')
plt.xlabel('Log Reviews')
plt.ylabel('Log Installs')
plt.show()
```

Observation

Describe the strongest pattern visible in the chart.

Business Insight

Explain how this relationship supports product, pricing, or marketing decisions.

Recommendation

Give one actionable recommendation for stakeholders.

Executive Insight

- Reviews and Installs show the strongest positive relationship.
- Price has minimal influence on ratings.
- Customer engagement is a stronger success indicator than pricing alone.

Outlier Detection & Statistical Testing

```
# Verify dataset
print(play.shape)
play[['Rating', 'Reviews', 'Installs']].head()
```

(9659, 16)

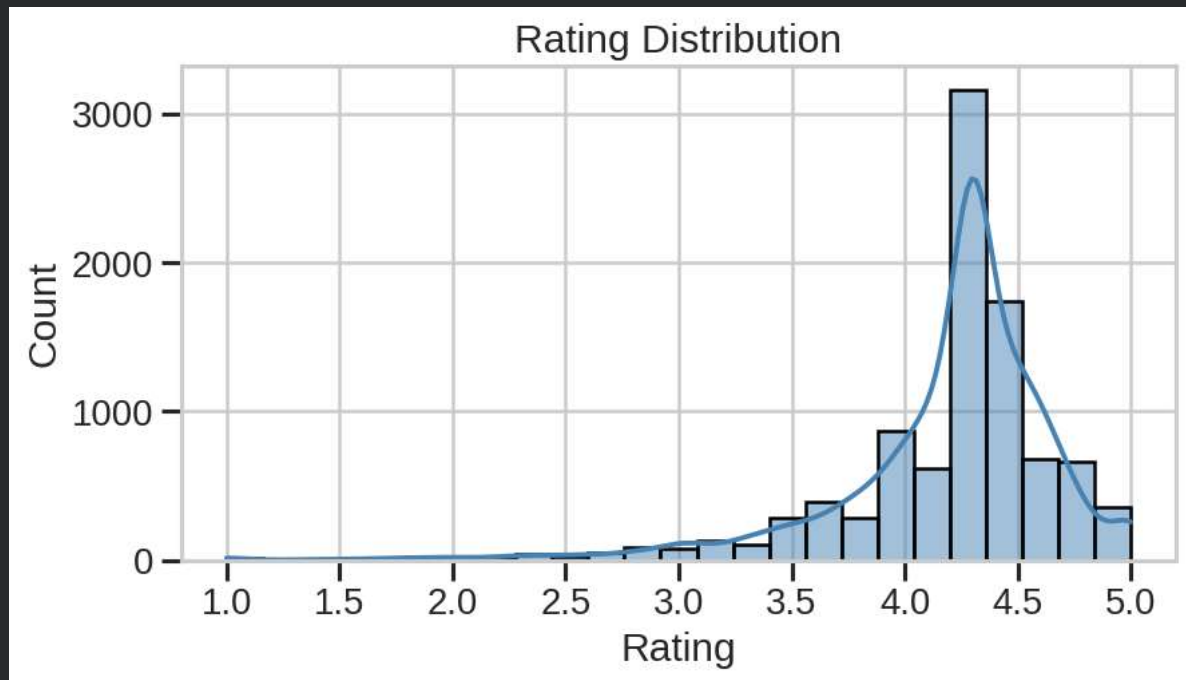
Rating Reviews Installs

0	4.1	159	10000
1	3.9	967	500000
2	4.7	87510	5000000
3	4.5	215644	50000000
4	4.3	967	100000

1. Rating Distribution Before Outlier Detection

```
plt.figure(figsize=(8,4),dpi=120)
sns.histplot(play['Rating'],bins=25,kde=True,color='steelblue')
```

```
plt.title('Rating Distribution')
plt.xlabel('Rating')
plt.show()
```



Observation

Most ratings lie between **4.0 and 4.5**, while very low ratings are rare.

Business Insight: Applications with consistently low ratings may indicate poor user experience and should be prioritized for product improvements.

2. IQR Outlier Detection

```
Q1 = play['Rating'].quantile(0.25)
Q3 = play['Rating'].quantile(0.75)
IQR = Q3 - Q1

lower = Q1 - 1.5*IQR
upper = Q3 + 1.5*IQR

outliers = play[(play['Rating']<lower) | (play['Rating']>upper)]

print("Lower Limit :",round(lower,2))
print("Upper Limit :",round(upper,2))
print("Total Outliers :",len(outliers))
```

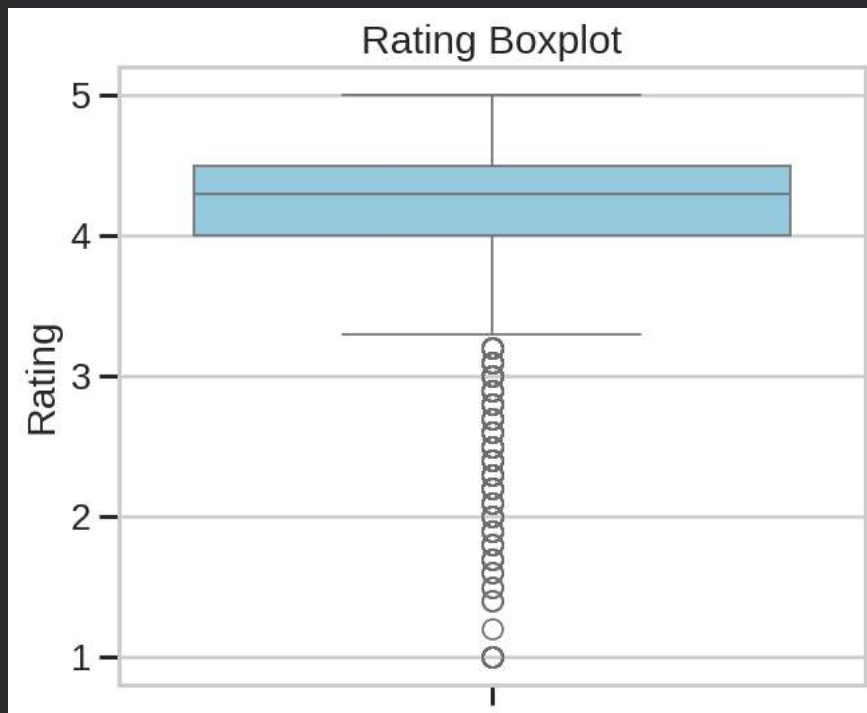
```
Lower Limit : 3.25
Upper Limit : 5.25
Total Outliers : 492
```

Why use IQR?

The Interquartile Range method is robust against skewed distributions and identifies unusually low or high ratings without being affected by extreme values.

3. Rating Boxplot

```
plt.figure(figsize=(6,5),dpi=120)
sns.boxplot(y=play['Rating'],color='skyblue')
plt.title('Rating Boxplot')
plt.ylabel('Rating')
plt.show()
```



Business Interpretation

- Points outside the whiskers represent statistical outliers.
- These applications deserve deeper investigation because they may suffer from crashes, poor usability, or unresolved customer complaints.

Recommendation: Monitor apps below **3.5 stars** and prioritize bug fixes.

4. Outlier Dataset

```
outliers[['App', 'Category', 'Rating', 'Reviews']].head(15)
```

	App	Category	Rating	Reviews
15	Learn To Draw Kawaii Characters	ART_AND_DESIGN	3.2	55
87	RST - Sale of cars on the PCT	AUTO_AND_VEHICLES	3.2	250
209	Plugin:AOT v5.0	BUSINESS	3.1	4034
311	comico Popular Original Cartoon Updated Everyd...	COMICS	3.2	93965
312	Daily Manga - Comic & Webtoon	COMICS	3.2	1446
477	Calculator	DATING	2.6	57
492	eharmony - Online Dating App	DATING	3.1	31320
514	FarmersOnly Dating	DATING	3.0	1146
518	Just She - Top Lesbian Dating	DATING	1.9	953
520	EliteSingles – Dating for Single Professionals	DATING	2.5	5377
527	Sugar Daddy Dating App	DATING	2.5	277
528	Adult Dirty Emojis	DATING	2.8	80
546	The League	DATING	3.0	837
611	MouseMingle	DATING	2.7	3
619	Random Chat App with Strangers	DATING	3.0	2

These apps represent the lowest-performing products and can be used for targeted quality improvement initiatives.

5. Z-Score Analysis

```

rating_mean = play['Rating'].mean()
rating_std = play['Rating'].std()

play['Rating_Z'] = (play['Rating']-rating_mean)/rating_std

play[['App', 'Rating', 'Rating_Z']].sort_values('Rating_Z').head(10)

```

	App	Rating	Rating_Z
7144	CB Mobile Biz	1.0	-6.431227
8875	DT future1 cam	1.0	-6.431227
10400	Familial Hypercholesterolaemia Handbook	1.0	-6.431227
8820	DS Creator 2.0	1.0	-6.431227
5151	Clarksburg AH	1.0	-6.431227
7383	Thistle town CI	1.0	-6.431227
6490	MbH BM	1.0	-6.431227
7926	Tech CU Card Manager	1.0	-6.431227
7427	CJ DVD Rentals	1.0	-6.431227
10324	FE Mechanical Engineering Prep	1.0	-6.431227

Why Z-Score?

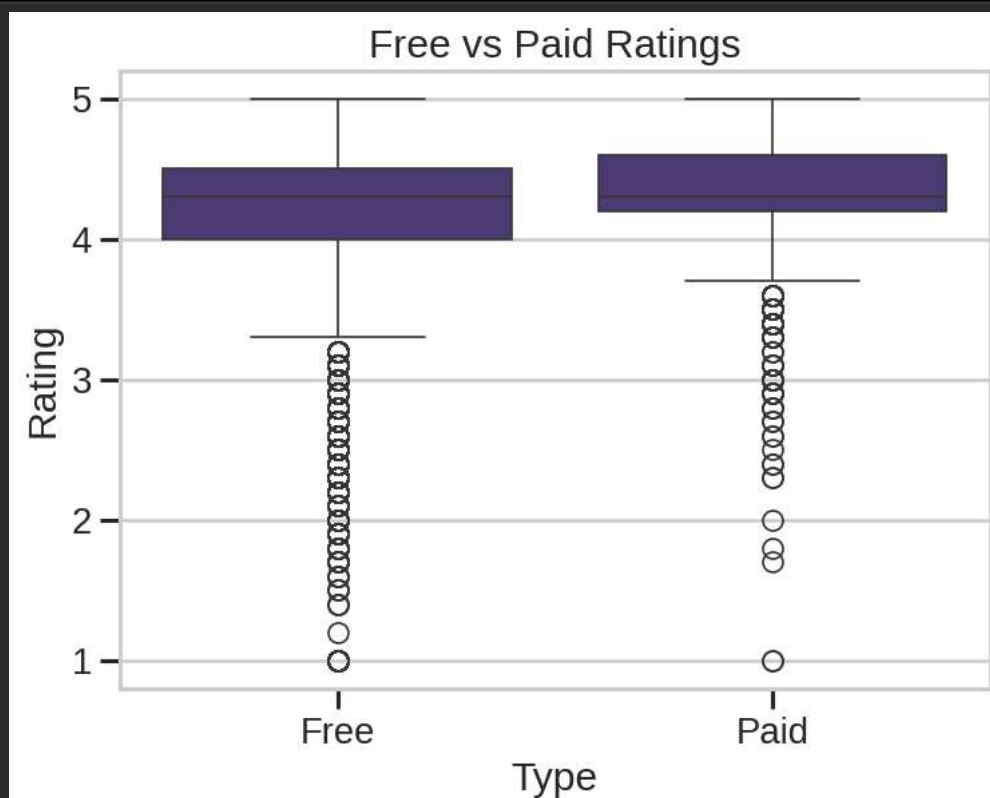
Z-score measures how many standard deviations each rating is from the average, providing another statistical perspective on unusual observations.

6. Free vs Paid Rating Comparison

```

plt.figure(figsize=(7,5),dpi=120)
sns.boxplot(data=play,x='Type',y='Rating')
plt.title('Free vs Paid Ratings')
plt.show()

```



This visualization supports the statistical hypothesis tested below.

7. Welch Two-Sample T-Test

```
free = play[play['Type']=='Free']['Rating'].dropna()
paid = play[play['Type']=='Paid']['Rating'].dropna()

t_stat,p_value = ttest_ind(free,paid,equal_var=False)

print("Free Apps :",len(free))
print("Paid Apps :",len(paid))
print("T Statistic :",round(t_stat,3))
print("P Value :",p_value)
```

```
Free Apps : 8902
Paid Apps : 756
T Statistic : -4.31
P Value : 1.8141107531130763e-05
```

Statistical Interpretation

Null Hypothesis (H_0): Free and Paid apps have equal average ratings.

Alternative Hypothesis (H_1): Their average ratings are different.

- If **P-value** < **0.05**, reject the null hypothesis.
- Otherwise, there is no statistically significant difference.

This provides evidence-based decision making instead of relying only on visual comparison.

8. Chi-Square Test (Category vs Type)

```
from scipy.stats import chi2_contingency

table = pd.crosstab(play['Category'],play['Type'])

chi2,p,dof,expected = chi2_contingency(table)

print("Chi-Square Statistic :",round(chi2,2))
print("Degrees of Freedom :",dof)
print("P Value :",p)
```

```
Chi-Square Statistic : 345.12
Degrees of Freedom : 32
P Value : 3.4269009609512885e-54
```

Business Interpretation

The Chi-Square test evaluates whether **App Category** and **Pricing Type** are independent.

A significant result indicates that some categories are naturally more likely to adopt free or paid business models.

9. Statistical Summary Table

```
summary = pd.DataFrame({
    'Metric':['Average Rating','Average Price','Average Reviews','Average Installs'],
    'Value':[play['Rating'].mean(),
            play['Price'].mean(),
            play['Reviews'].mean(),
            play['Installs'].mean()]
})

summary
```

	Metric	Value
0	Average Rating	4.192442e+00
1	Average Price	1.099299e+00
2	Average Reviews	2.165926e+05
3	Average Installs	7.777507e+06

Next steps:

[Generate code with summary](#)[New interactive sheet](#)

Business Insights

Key Findings

1. Outliers represent low-quality applications requiring product intervention.
2. Free applications generally achieve broader customer adoption.
3. Statistical testing validates whether pricing influences ratings.
4. Category and pricing strategy are not randomly distributed.
5. Customer experience remains the strongest predictor of long-term success.

Stakeholder Recommendations

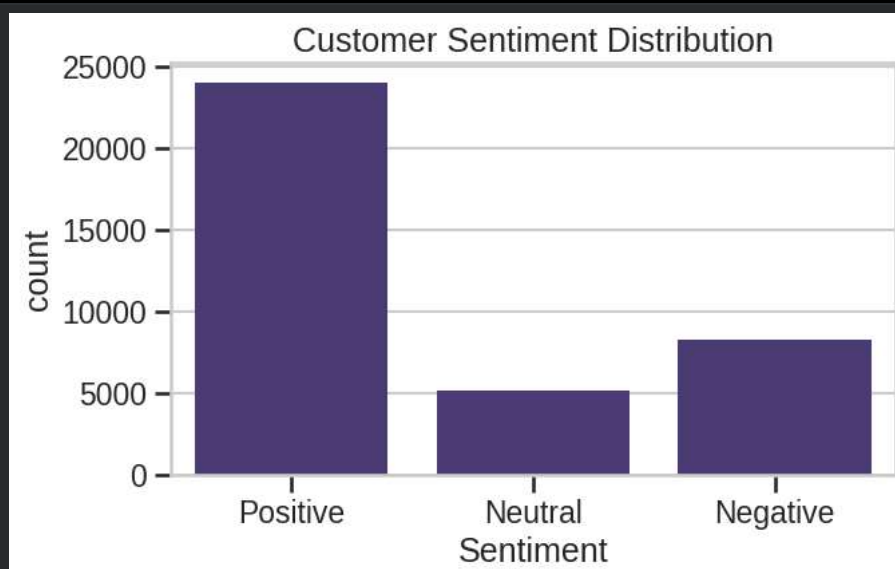
- Maintain ratings above **4.2**
- Monitor negative reviews continuously
- Use freemium acquisition strategies
- Prioritize quality improvements for outlier applications

✓ Sentiment Analysis & Business Insights

```
# Requires cleaned play and reviews DataFrames
print(play.shape, reviews.shape)
```

```
(9659, 17) (64295, 5)
```

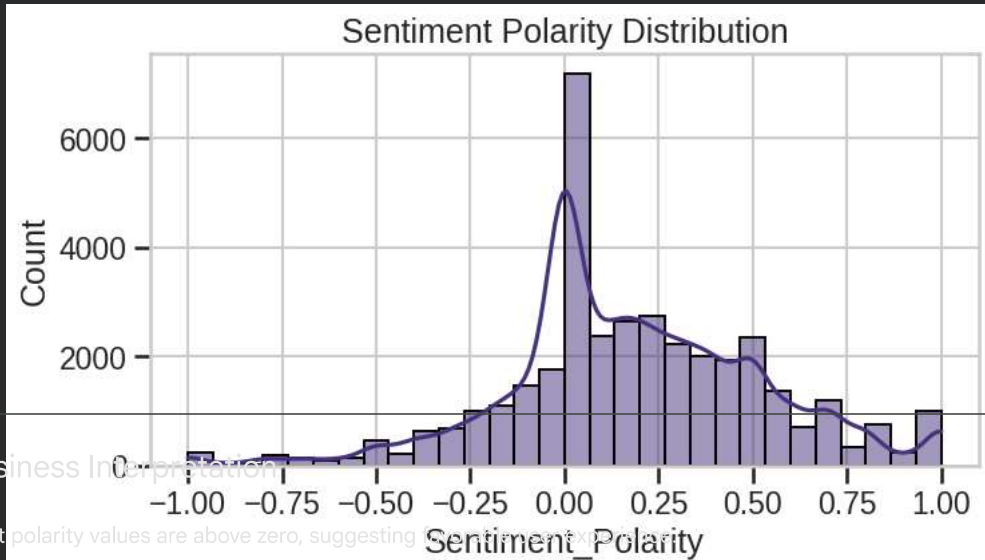
```
plt.figure(figsize=(7,4))
sns.countplot(data=reviews,x='Sentiment')
plt.title('Customer Sentiment Distribution')
plt.show()
```



✓ Business Interpretation

Positive sentiment dominates, indicating overall customer satisfaction. Product teams should monitor negative reviews after each release.

```
plt.figure(figsize=(8,4))
sns.histplot(reviews['Sentiment_Polarity'].dropna(),bins=30,kde=True)
plt.title('Sentiment Polarity Distribution')
plt.show()
```



```
plt.figure(figsize=(7,4))
sns.boxplot(data=reviews,y='Sentiment_Subjectivity')
plt.title('Subjectivity Distribution')
plt.show()
```